SOFTWARE ARCHITECTURE

Distributed Computing

BEYOND BOUNDARIES

HOA SEN UNIVERSITY
WORLD CLASS EDUCATION

QS STARS RATING SYSTEM

ACBSP ACCREDITED

AUN-QA

MOET ACCREDITED
KIỂM ĐỊNH CHẤT LƯỢNG GIÁO DỤC QUỐC GIA

Definition

- Distributed computing is an art of computing to interact with a collection of independent systems in the field of computer science.
- Core characteristics
  - Concurrency
  - Scalability
  - Transparency
  - Security
  - and so on.

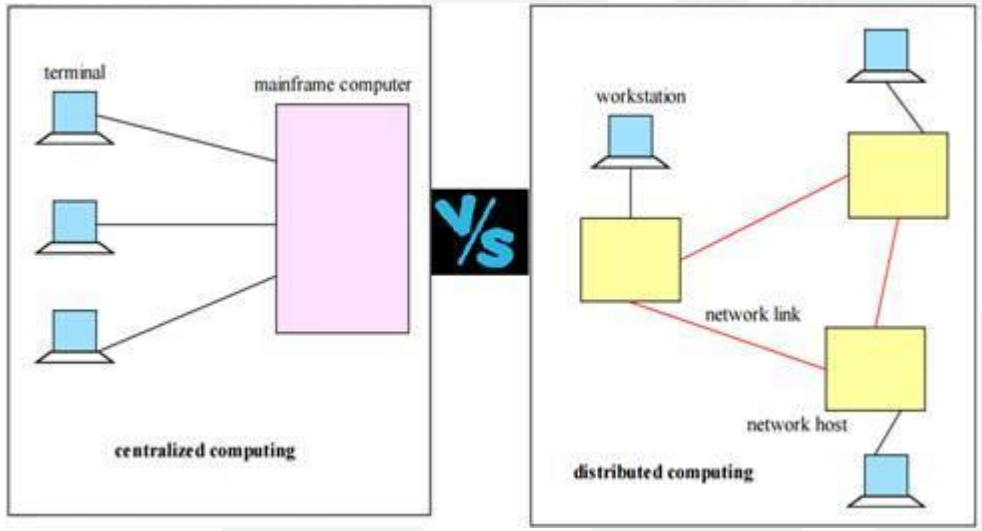
Definition of the computing

- In the early days of computer evolution, mainframe computers were heavily used in the industry.
- By design, the legacy mainframe was performed on a single system and was called centralized computing

This Photo by Unknown Author is licensed under CC BY-NC-ND

Distributed system

- A distributed system is a collection of independent computers, interconnected via a network.
- The core objective of the network is to collaborate and complete a particular task.
- Fundamentally, distributed computing is computing performed in a distributed system



Distributed systems strong points

- The availability of powerful
- Cheaper microprocessors in personal computers, embedded systems, personal digital assistant

Comparison

Sr.No.	Centralized	Distributed
1	Monolithic architecture within a single system.	Distributed architecture across the network.
2	In terms of functionality, separate and single function applications are designed.	Integrated applications are designed in terms of functionality.
3	Applications can't share data or other app resources by design.	By design, applications can share all types of resources.
4	Mainframe-based applications are developed using this approach.	Modern multi-layer/tier applications are designed based on this methodology.
5	Proprietary user interfaces are developed here.	Common user interfaces are designed here.

Discussion

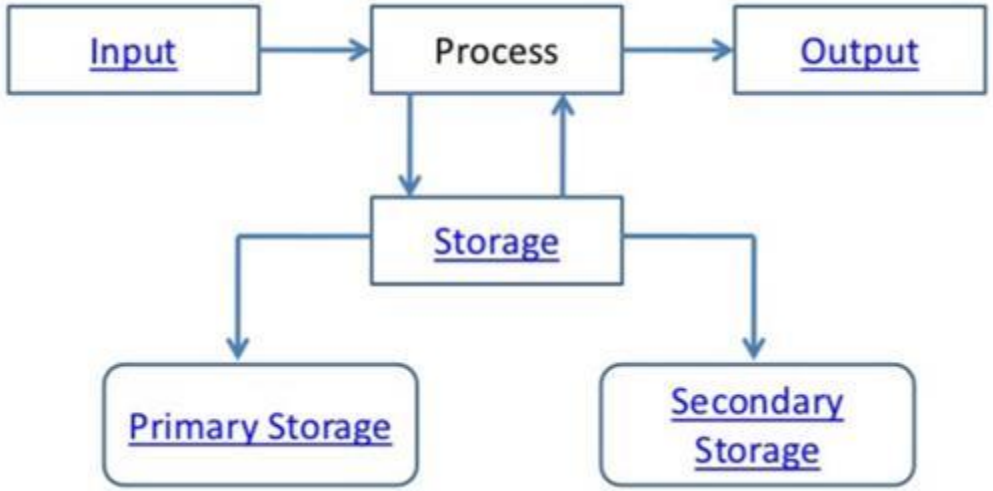
- Let's read the article Centralized Computing vs. Distributed Computing at https://www.baeldung.com/cs/centralized-vs-distributed-computing
- In your opinion, in which kind of software/ application, Centralized Computing is a suitable choice, and in what situation, Distributed Computing is a good choice?

Multiprogramming

- Since the advent of the Internet, new enterprise applications being built on distributed processing.
- The technical journey goes in the way of intranet, Internet, World Wide Web, wireless network, GPS, RFID (Radio Frequency Identifier), and so on.
- In the emerging world of global computing, distributed computing is the core central piece of all computing.

Multiprogramming

- Four core components:
  - Input
  - Process
  - Storage
  - Output



Multiprogramming

- The **Input** phase ingests the data and commands into CPU using one or more input devices.
- The **Process** phase performs the execution of the input data and turns it into useful information as **Output**.
- During this process, it persists the useful data into the **Storage** layer.

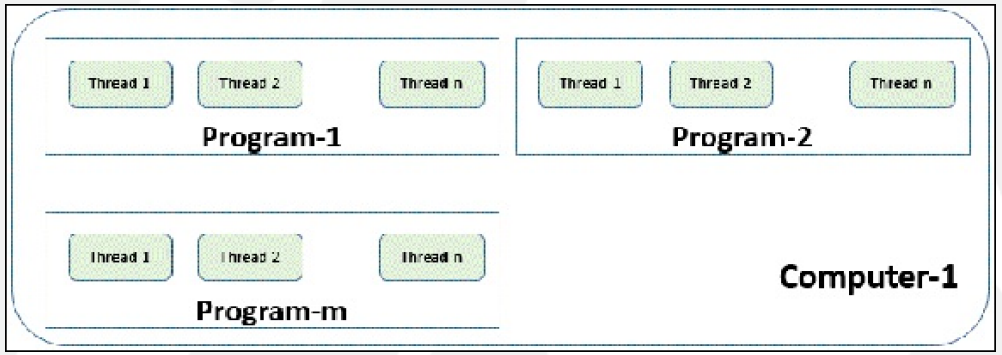
Multiprogramming vs multithreading

- Multiprogramming is the capability of an operating system to execute **more than one program** on a single machine.
  - A classic example is to run more than one program, such as Excel, Internet Explorer, Word, and more, in a Windows operating system:
- Multithreading has the ability of an operating system to execute **different parts of a program**, namely--threads.

Multiprogramming vs multithreading

- Multiprogramming

- Multithreading

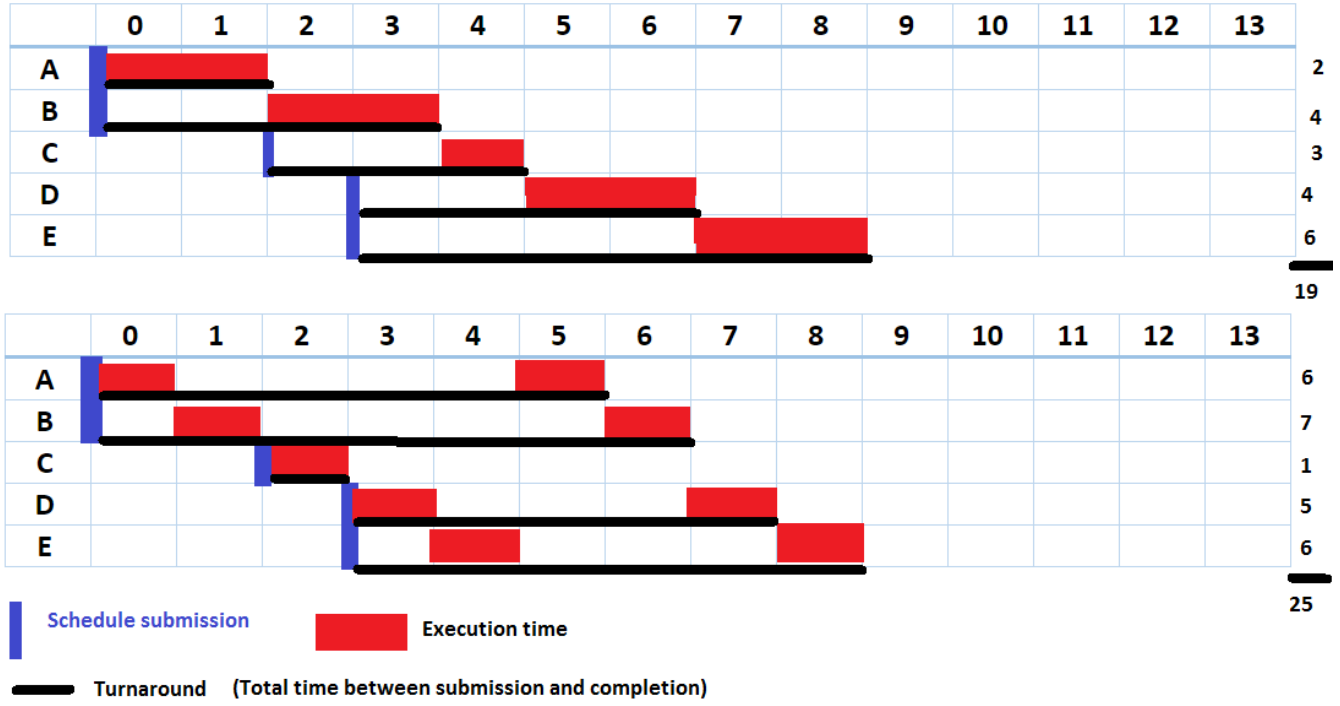


Threads in multithreading

- Threads are considered as child processes to share the parent process resources but execute independently.
- In terms of parallel execution, multiple threads of a single process **shares the single CPU system** on time-sliced basis.
- A classic example is a multiple document process using a single Word process.

Thread synchronization

- Using Time Slice mechanism to switch between different threads.
- It is based on round-robin methodology.



This Photo by Unknown Author is licensed under CC BY-SA

Process-based vs Thread-based multitasking

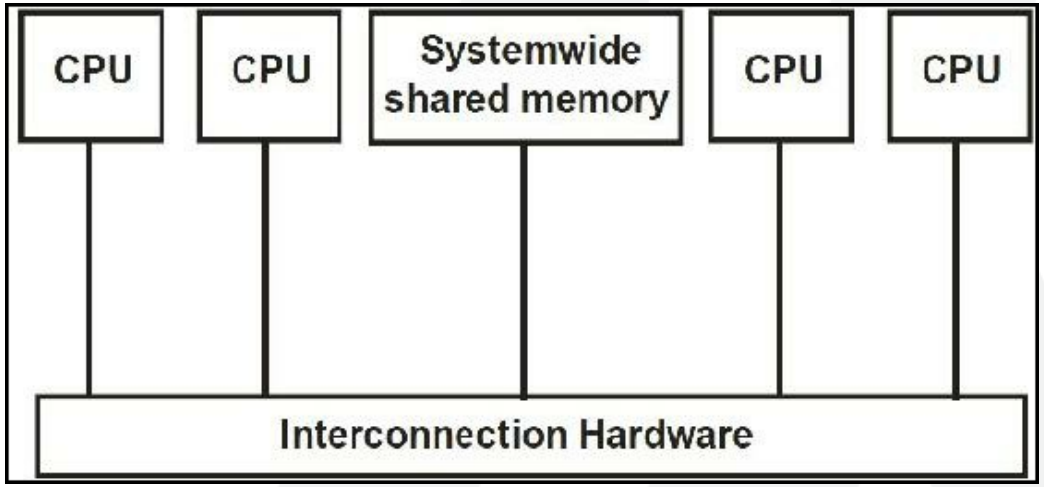
- Process-based multitasking handles the concurrent execution of programs
- Thread-based multitasking deals with the concurrent execution of pieces of the same program.

Storage

- Computer architecture is designed with the interconnected multiple processors types in two forms:
  - Tightly (highly) coupled system
  - Loosely coupled system

Storage

- Tightly (highly) coupled system
  - A single system-wide primary memory, which is shared by all the processors in the ecosystem.
  - Any sort of communication is executed in the closed tightly couple environment of the common shared memory route



Storage

- Tightly (highly) coupled system example
  - The multiprocessor environment.
  - Processor 1 writes value 100 into the shared memory address location ADDR1.
  - Subsequent reads by the rest of the processors will be the same value, 100, from the ADDR1 location.
  - It will be valid until there are any further updates on the specific location.

Storage

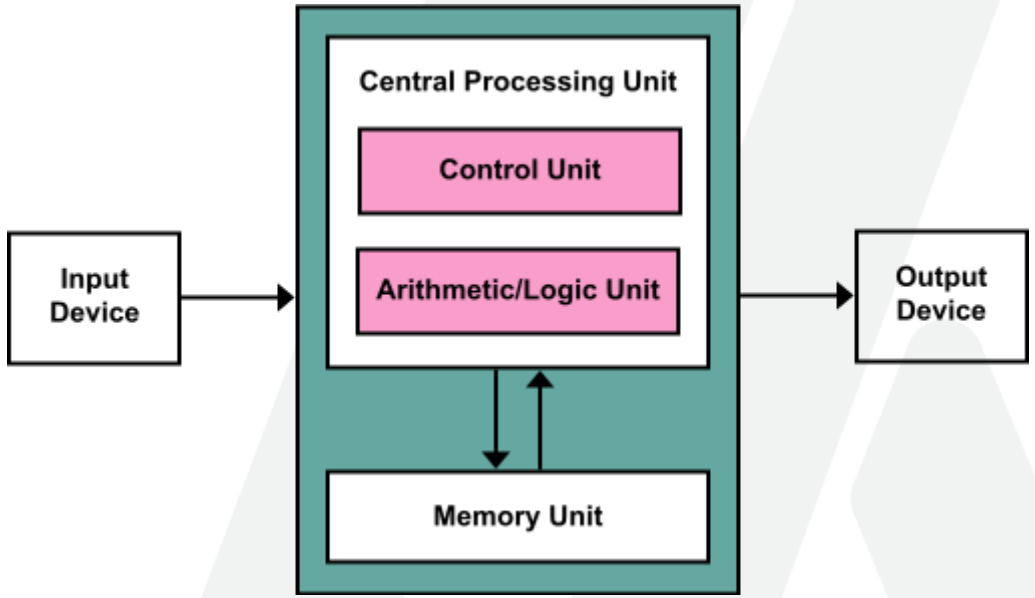
- Loosely coupled system
  - Each processor contains its own independent local memory to persist the value.
  - It doesn't share any memory location across the processors.
  - By architecture, this is termed Message Driven Architecture in the computing industry.

Coupled system and distributed computing.

- The first version tightly coupled system is built using **parallel computing.**
- The second version Loosely Coupled system is based on **distributed computing.**

Process

- The computing process is defined as an instance of a software program to execute a **set of predefined activities to generate the result** as the output.
- In the hardware of the computers, CPU is the core component of the process in which Control Unit (CU) and ALU (Arithmetic Logic Unit) are the fundamentals to execute.



This Photo by Unknown Author is licensed under CC BY-SA-NC

Process performance

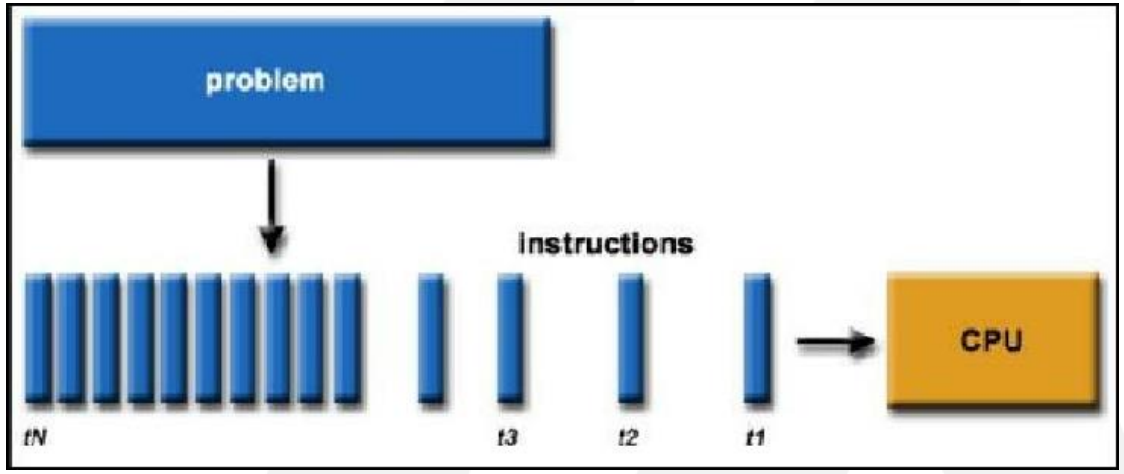
- Performance is the key success indicator of the process
- Two aspects to a performance:
  - The execute multiples of a single job instance.
  - The multiple folds at the machine level.

Concurrency

- The program is decomposed into multiple threads to execute the control with distinct responsibilities.
- Multithreads may run either simultaneously or in turn based on the system infrastructure.
- Two main concepts:
  - How are the processes scheduled to execute?
  - How does the coordination process synchronize to produce the result?

Concurrency example

- The given business problem statement is converted into a software program.
- Multiple instructions of the program are decomposed into logical independent splits, named Thread.
- Therefore, the program/problem 'p' is split into 't' threads (t1, t2,...tN), the execution target at one CPU processor.



Concurrency

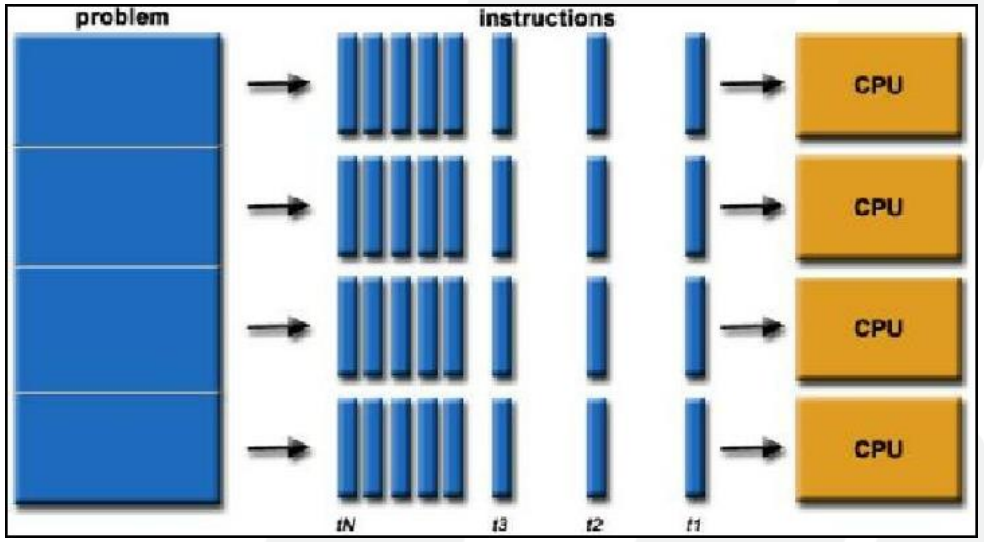
- Concurrency systems have several processes that are executing simultaneously.
- These processes can interact with each other while they are executing.
- Concurrent programming is implemented by splitting the existing task into multiple factors to be executed.
- Concisely, concurrency achieves logical execution of multiple tasks at once.

Parallelism

- Parallel computing has a major design difference with the earlier concurrent computing.
- It reduces the execution cycle/timing by taking advantage of the infrastructure/hardware ability to execute/complete more than one task at any point of given time.
- Parallelism is a combination of software and hardware techniques.
- It is used to allow several processes to run in parallel, physically.
- The primary objective is used for faster computation in modern enterprise applications.

Parallelism

- Parallelism achieves the physical execution of multiple tasks at once using multiple processors in the ecosystem.



Q&A

This Photo by Unknown Author is licensed under CC BY-NC-ND